

Notes: Geczy, Minton, and Schrand (JF, 1997)

Why Firms Use Currency Derivatives

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1 Big picture

- **Question.** Why do large nonfinancial firms use currency derivatives, and which theory of corporate hedging best explains cross-sectional variation in derivative use?
- **Setting.** The paper studies 372 Fortune 500 nonfinancial firms with ex ante foreign exchange exposure in fiscal year 1990/1991.
- **Main empirical object.** The dependent variable is a disclosure-based indicator for whether the firm uses currency derivatives: swaps, forwards, futures, options, or combinations of these instruments.
- **Main finding.** Firms with greater growth opportunities and tighter financial constraints are more likely to use currency derivatives.
- **Interpretation.** The evidence supports underinvestment/cash-flow volatility theories of hedging: firms hedge to reduce cash-flow variation that could otherwise force them to pass up valuable investment opportunities.
- **Exposure finding.** Firms with greater foreign exchange exposure, measured using foreign income, foreign sales, foreign-denominated debt, and import competition, are more likely to use currency derivatives.
- **Cost finding.** Larger firms and firms that already use other derivatives are more likely to use currency derivatives, consistent with fixed costs and economies of scale in hedging programs.
- **Natural hedge result.** Foreign-denominated debt and currency derivatives appear to substitute for each other when hedging foreign operations.
- **Instrument-choice result.** The type of foreign exchange exposure matters for the choice between swaps and forwards/options. Operating exposure and financing exposure do not map into the same hedging instruments.
- **Takeaway.** Currency derivatives use is not random. It is systematically related to the benefits and costs of hedging, especially investment opportunities, financial constraints, exposure, and scale economies.

2 Incremental contribution of the paper

2.1 Contribution relative to hedging theory

- Earlier theoretical work offers multiple motives for hedging: managerial risk aversion, distress costs, underinvestment problems, taxes, agency costs of debt, and imperfect capital markets.
- This paper organizes these theories into a common empirical framework and maps each theory into observable firm characteristics.
- The paper emphasizes that a theory of hedging must explain not only whether firms face risk, but whether hedging benefits exceed hedging costs.

Interpretation: The paper moves the literature from broad arguments that derivatives may be useful to a structured empirical horse race among competing theories.

2.2 Contribution relative to early empirical derivatives studies

- Earlier evidence often relied on survey data or broad samples that included many firms with little reason to hedge currency risk.
- This paper uses annual-report disclosures, which became available after accounting disclosure changes, to construct firm-level derivative-use measures.

- It restricts attention to firms with ex ante foreign exchange-rate exposure, reducing noise from firms that have no meaningful currency risk.
- It retains broad cross-industry variation, unlike single-industry hedging studies.

Interpretation: The sample design is a major contribution: it combines broad cross-sectional variation with an exposure screen that makes the hedging decision economically meaningful.

2.3 Contribution relative to speculation concerns

- A common concern is that derivative use may reflect speculation rather than hedging.
- The paper directly discusses how speculative motives would affect the empirical predictions.
- Many of the key variables supporting the main findings, especially growth opportunities and financial constraints, are natural hedging motives but not natural speculation motives.

Interpretation: The paper cannot observe the exact hedge position or risk reduction, but the pattern of coefficients is much more consistent with hedging than with speculation.

2.4 Contribution on instrument choice

- The paper goes beyond the binary question of derivative use and studies which instruments firms choose.
- It distinguishes firms using swaps or swaps combined with other instruments from firms using forwards/options or combinations without swaps.
- This shows that the source of currency exposure matters for the structure of the hedging program.

Interpretation: Instrument choice evidence makes the paper richer than a simple users-versus-nonusers study. It shows that hedging is tailored to the form of exposure.

3 Data and measurement

3.1 Sample construction

- The initial population is Fortune 500 nonfinancial firms.
- The final sample contains 372 large U.S. industrial firms with foreign exchange exposure.
- A firm is classified as having exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of sample firms based on imports as a percentage of total industry sales.
- The focus on 1990/1991 reflects the availability of annual-report disclosures about derivative use.

Interpretation: The exposure screen is important. A firm without currency exposure may not use currency derivatives simply because it does not need them, not because hedging theories fail.

3.2 Dependent variables

- **Currency derivative user:** an indicator equal to one if the firm discloses the use of currency swaps, forwards, futures, options, or combinations.
- **Any derivative user:** an indicator equal to one if the firm uses currency, interest-rate, or commodity derivatives.
- **Instrument choice:** a multinomial category distinguishing no currency instrument, swaps/swap combinations, and forwards/forward combinations.

Interpretation: The main dependent variable measures disclosed use, not the notional size or delta of the hedge. The paper therefore studies the extensive margin of derivative use.

3.3 Hedging incentive variables

The paper maps theories into the following variables:

- **Managerial incentives:** managerial wealth and managerial option ownership.
- **Financial distress and bondholder incentives:** interest coverage, long-term debt ratio, and interactions with growth opportunities.
- **Underinvestment incentives:** R&D/sales, book-to-market, plant/property/equipment investment, and quick ratio.
- **Substitutes for hedging:** convertible debt, preferred stock, dividend yield, and tax-loss carryforwards.
- **Information and governance variables:** institutional ownership and analyst coverage.
- **Scale:** firm size and whether the firm uses other derivatives.

Interpretation: The paper’s strength is that the variables are explicitly connected to theory. This allows the regression evidence to distinguish among competing explanations.

3.4 Foreign exchange exposure measures

- Foreign pretax income scaled by total sales.
- Foreign sales and exports scaled by total sales.
- Identifiable foreign assets scaled by total assets.
- Foreign long-term debt and foreign short-term debt scaled by total assets.
- Imports as a share of four-digit SIC industry sales.

Interpretation: These variables capture operating, financing, and competitive sources of currency exposure. The paper uses them both to predict derivative use and to explain instrument choice.

4 Empirical specifications

4.1 Binary logit model of currency derivatives use

The main regression is a logit model:

$$P(Use_i = 1 \mid X_i) = \Lambda(\alpha + X_i' \beta), \quad (1)$$

where Use_i indicates currency derivative use and X_i includes hedging incentives, exposure variables, industry indicators, and size.

The paper reports marginal effects:

$$\Delta Prob = \frac{\partial P(Use_i = 1 \mid X_i)}{\partial X_{ki}}, \quad (2)$$

computed at the mean of the regressors.

Interpretation: The marginal effects show how a one-unit change in a variable changes the probability of currency derivative use.

4.2 Full sample and partial sample

- The full sample regression includes 282 firms with available book-to-market data.
- The partial sample includes 220 firms for which R&D expense data are available.
- The partial sample tests underinvestment predictions more directly because R&D is a proxy for growth opportunities.

Interpretation: Book-to-market and R&D are alternative proxies for growth opportunities, and the paper checks that the interpretation is robust across both.

4.3 Robustness and endogeneity checks

The paper addresses the concern that capital structure, managerial compensation, and derivative use may be jointly chosen. It estimates alternative models using residualized measures and alternative proxies for leverage and managerial incentives.

Interpretation: The authors cannot fully solve all simultaneity concerns, but they show the main findings are not sensitive to obvious alternative specifications.

4.4 Multinomial logit model of instrument choice

To study instrument choice, the paper estimates a multinomial logit model:

$$P(\text{Choice}_i = j \mid X_i) = \frac{\exp(X_i' \beta_j)}{1 + \sum_{k=1}^2 \exp(X_i' \beta_k)}, \quad (3)$$

where the baseline category is no currency instrument and the alternatives are swaps/swap combinations and forwards/forward combinations.

Interpretation: The multinomial model tests whether different exposure sources lead firms to use different instruments, not merely whether they hedge at all.

5 Identification and empirical interpretation

5.1 What identifies the main result

- Identification is cross-sectional.
- The central comparison is between exposed firms that use currency derivatives and exposed firms that do not.
- The key assumption is that observed proxies for incentives and exposure capture meaningful differences in benefits and costs of hedging.
- Industry indicators help absorb broad sectoral differences in derivative use.

Interpretation: This is not a causal design in the modern quasi-experimental sense. It is a theory-driven cross-sectional test.

5.2 Why the exposure screen matters

- Without an exposure screen, nonuse could mean no need to hedge.
- With the screen, both users and nonusers have plausible foreign currency exposure.
- This makes differences in derivative use more informative about hedging motives and hedging costs.

Interpretation: The paper's empirical strategy is to hold the existence of exposure roughly constant and explain variation in whether firms hedge that exposure.

5.3 Main limitations

- The data measure derivative use, not exact hedging intensity or hedge effectiveness.
- Annual-report disclosures may be incomplete or imprecise.
- Some firms could use derivatives for speculation or selective hedging.
- The empirical design is cross-sectional and cannot fully address endogeneity.
- The sample is large Fortune 500 firms, so the conclusions may not apply to small firms.

Interpretation: The paper's value is not definitive causal identification. Its value is careful theory mapping and early systematic evidence on why large firms use currency derivatives.

6 Tables and figures

6.1 Table I and Table II: predicted signs and use frequencies

Read:

- Table I maps theory to predicted signs for variables used in the empirical tests.
- It distinguishes managerial, bondholder, and shareholder theories of hedging.
- Table II shows that 41.4% of exposed Fortune 500 firms use currency derivatives, while 59.1% use some derivative.
- Use rises sharply with size: 75.3% of firms in the largest sales quartile use currency derivatives, compared with 17.2% in the smallest quartile.
- Currency derivative use varies by industry, with high rates in consumer goods, electronics, production materials, and transportation.

Economic message: Table I defines the empirical horse race. Table II shows that derivative use is common but far from universal, leaving meaningful variation to explain.

Predicted signs of coefficient estimates for variables used as proxies for incentives to use foreign currency derivatives based on the testable implications of Smith and Stulz (1985, S&S), Froot, Scharfstein, and Stein (1993, FSS), and DeMarzo and Duffie (1991, D&D). "D" refers to a direct prediction of a model; "I" refers to an indirect prediction. "NA" indicates that the independent variable is not applicable to a particular model. "?" indicates that the predicted sign is indeterminate. Variable names appear in the second column. Predicted signs appear in the third through fifth columns.				
Independent Variables	Variable Name	Prediction		
		S&S (1985)	FSS (1993)	D&D (1991)
Managers				
Managerial wealth	MGRWLTH	+, D	NA	NA
Managerial option ownership	OPTS	-, D	NA	NA
Bondholders				
Interest coverage ratio	COV_AV	-, D	-, D	NA
Long-term debt ratio	DE_AV	+, D	+, D	NA
R&D expenses/sales	RD	+, I	+, D	+, I
Plant, property, and equipment investment expenditures/SIZE	PPE	+, I	+, D	+, I
Book-to-market ratio	BM	-, I	-, D	-, I
Substitutes for off-balance sheet assets				
Convertible debt/SIZE	CONV	-, I	+, I	NA
Preferred stock/SIZE	PS	-, I	+, I	NA
Quick ratio	QUICK_AV	-, I	-, D	NA
Dividend yield	DIV_AV	+, I	+, I	NA
Equityholders				
Tax-loss carryforwards/total assets	NOL	+, D	-, I	NA
Institutional ownership	IO	NA	NA	-, D
Number of analyst firms	ANLST	NA	NA	-, D

Table I: predicted signs

Frequency of use of derivative instruments by 372 large U.S. firms for fiscal year-end 1991 that have foreign exchange rate exposure as of fiscal year-end 1990. Companies are among the 500 largest firms (by sales) in the Fortune 500. A firm has foreign exchange rate exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firms on the basis of imports as a percentage of total industry sales. Currency Derivatives include currency swaps and foreign exchange forwards, futures, and options. Any Derivatives include interest rate, commodity, and currency derivatives. All data on derivatives use are from annual reports and 10-K disclosures. The 1st quartile for firm size includes the smallest firms based on 1990 sales; the 4th quartile includes the largest firms.			
	N	Currency Derivatives	Any Derivatives
<i>All Firms</i>	372	41.4%	59.1%
Panel A: By Firm Size (by 1990 sales)			
4th quartile	93	75.3	90.3
3rd quartile	93	38.7	64.5
2nd quartile	93	34.4	48.4
1st quartile	93	17.2	33.3
Panel B: By Fortune's Industry Grouping			
<i>Consumer Goods</i>	47	66.0%	78.7%
Beverages	6	83.3	100.0
Food	22	59.1	81.8
Pharmaceuticals	14	85.7	85.7
Tobacco	5	20.0	20.0
<i>Electronics</i>	71	56.3%	63.4%
Computers, office equipment	18	83.3	88.9
Electronics, electrical equipment	35	42.9	48.6
Scientific, photographic, and control equipment	18	55.6	66.7
<i>Energy</i>	32	34.4%	68.8%
Mining, crude oil production	12	8.3	58.3
Petroleum refining	20	50.0	75.0
<i>Metals</i>	32	21.9%	50.0%
Jewelry, silverware	1	0.0	0.0
Metal products	19	21.1	47.4
Metals	12	25.0	58.3
<i>Nondurable consumer products</i>	35	28.6%	42.8%
Apparel	11	27.3	36.4
Furniture	5	0.0	20.0
Soaps, cosmetics	11	36.4	36.4
Textiles	6	16.7	16.7
Toys, sporting goods	2	100.0	100.0
<i>Paper</i>	41	17.1%	39.0%
Forest and paper products	27	18.5	44.4
Publishing, printing	14	14.3	28.6
<i>Production materials</i>	50	44.0%	62.0%
Building materials, glass	7	57.1	100.0
Chemicals	33	42.4	57.6

Table II: derivative use by size and industry

6.2 Table III: users versus nonusers

Read:

- Panel A compares hedging incentives across currency derivative users and nonusers.
- Users have higher managerial option ownership, higher R&D/sales, more institutional ownership, more analyst coverage, and larger firm size.
- Users have lower book-to-market ratios, consistent with more growth opportunities.

- Panel B compares exposure variables. Users have more foreign income, foreign sales/export exposure, foreign long-term debt, and import competition exposure.

Economic message: The univariate evidence already supports the two central mechanisms: derivative users have both stronger incentives to hedge and greater currency exposure.

Table III
Summary of Financial Characteristics of Currency Derivative Users and Nonusers

Selected summary statistics for managerial and financial characteristics for firms that disclose the use of currency derivative instruments (currency derivative users), and firms that do not (currency derivative nonusers) for 372 large U.S. firms with foreign exchange rate exposure. Companies are among the 500 largest firms (by sales) in the Fortune 500. A firm has foreign exchange rate exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firms on the basis of imports as a percentage of total industry sales. Panel A reports summary statistics for proxies related to incentives for hedging. Panel B reports summary statistics for proxies for foreign exchange exposure. All data are measured as of fiscal year-ends. In the cases of missing data, the number of observations are given in parentheses. The *t*-statistics are given for tests of the equality of means between currency derivative users and nonusers. The *p*-values are in parentheses.

Selected summary statistic for managerial and financial characteristics for firms that disclose the use of currency derivative instruments (currency derivative users), and firms that do not (currency derivative nonusers) for 372 large U.S. firms with foreign exchange rate exposure. Companies are among the 500 largest firms (by sales) in the Fortune 500. A firm has foreign exchange rate exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firm on the basis of imports as a percentage of total industry sales. Panel A reports summary statistics for proxies related to incentives for hedging. Panel B reports summary statistics for proxies for foreign exchange exposure. All data are measured as of fiscal year-ends. In the cases of missing data, the number of observations are given in parentheses. The *t*-statistics are given for tests of the equality of means between currency derivative users and nonusers. The *p*-values are in parentheses.

Variable	Currency Derivative Users (<i>N</i> = 154)			Currency Derivative Non-Users (<i>N</i> = 218)			<i>t</i> -statistic ^a (<i>p</i> -value)
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	
Panel A: Incentives for Hedging							
Managerial wealth (log \$MM)	\$ 8.23 (<i>N</i> = 141)	\$ 8.02	\$ 1.64	\$ 7.93 (<i>N</i> = 203)	\$ 7.86	\$ 1.79	-1.60 (0.11)
Managerial option ownership (log \$MM)	7.45 (<i>N</i> = 131)	7.60	1.47	6.46 (<i>N</i> = 169)	6.50	1.37	-5.97 (0.00)
Interest coverage ratio	7.96 (<i>N</i> = 152)	4.43	17.00	10.56 (<i>N</i> = 213)	3.82	41.78	0.819 (0.41)
Long-term debt ratio	0.28	0.22	0.20	0.31 (<i>N</i> = 214)	0.31	0.20	1.71 (0.09)
R&D expenses/sales	0.05 (<i>N</i> = 136)	0.03	0.04	0.02 (<i>N</i> = 133)	0.02	0.02	-5.58 (0.00)
Plant, property, and equipment investment expenditures/SIZE	0.35	0.29	0.23	0.41	0.34	0.28	2.24 (0.03)
Book-to-market ratio	0.52	0.51	0.53	0.70	0.61	0.56	3.20 (0.00)
Convertible debt/SIZE	0.01 (<i>N</i> = 150)	0.00	0.00	0.02 (<i>N</i> = 216)	0.00	0.06	0.98 (0.33)
Preferred stock/SIZE	0.01	0.00	0.03	0.01	0.00	0.04	1.29 (0.20)
Quick ratio	0.25 (<i>N</i> = 153)	0.15	0.29	0.35	0.19	0.79	1.73 (0.08)
Dividend yield	0.03	0.03	0.09	0.04 (<i>N</i> = 214)	0.03	0.08	0.28 (0.78)
Tax-loss carryforwards/total assets	0.02 (<i>N</i> = 153)	0.00	0.08	0.02 (<i>N</i> = 213)	0.00	0.07	-0.26 (0.80)
Institutional ownership	55.46 (<i>N</i> = 152)	58.30	14.70	48.59 (<i>N</i> = 205)	52.00	16.67	-4.05 (0.00)
Number of analyst firms	26.16	25	13.14	15.43 (<i>N</i> = 211)	14	9.37	-8.65 (0.00)
Firm SIZE (log \$MM)	\$ 8.24	\$ 8.26	\$ 1.41	\$ 7.13 (<i>N</i> = 217)	\$ 7.13	\$ 1.01	-8.33 (0.00)

In addition to cross-sectional differences in hedging incentives, currency
Table III, Panel A: hedging incentives

Table III—Continued

Variable	Currency Derivative Users (<i>N</i> = 154)			Currency Derivative Non-Users (<i>N</i> = 218)			<i>t</i> -statistic (<i>p</i> -value)
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.	
Panel B: Foreign Exchange Exposure							
Pretax foreign income/total sales	0.033 (<i>N</i> = 149)	0.027	0.035	0.018 (<i>N</i> = 162)	0.008	0.039	-3.70 (0.00)
Foreign & export sales/total sales	0.389 (<i>N</i> = 141)	0.376	0.195	0.282 (<i>N</i> = 141)	0.228	0.241	-4.10 (0.00)
Identifiable foreign assets/total assets	0.346 (<i>N</i> = 138)	0.316	0.205	0.350 (<i>N</i> = 116)	0.240	0.313	0.12 (0.91)
Foreign long-term debt/total assets	0.023 (<i>N</i> = 153)	0.001	0.045	0.005 (<i>N</i> = 216)	0	0.014	-4.81 (0.00)
Foreign short-term debt/total assets	0.006 (<i>N</i> = 153)	0	0.018	0.003 (<i>N</i> = 216)	0	0.013	-1.85 (0.07)
Percentage of imports in direct sales	16.35%	12%	14.37%	12.03%	10%	11.05%	-3.13 (0.00)

Table III, Panel B: foreign exchange exposure

6.3 Table IV and Table V: logit model and correlations

Read:

- Table IV reports logit estimates of the probability of using currency derivatives.
- R&D/sales has a positive and significant marginal effect in the partial sample, supporting the underinvestment hypothesis.
- Quick ratio has a negative marginal effect, meaning internally liquid firms are less likely to use currency derivatives.
- Firm size, foreign income, foreign-denominated debt, and import competition are positive predictors of use.
- Table V reports correlations among explanatory variables, documenting potential multicollinearity and motivating robustness checks.

Economic message: The multivariate evidence shows that investment opportunities, financial constraints, scale, and exposure predict hedging even after controlling for other theories.

Table IV
Logit Regression Estimates of the Likelihood of Using Currency Derivatives

Logit regression estimates of the relation between the likelihood that a firm uses currency derivatives and proxies for incentives to use derivatives, proxies for foreign exchange exposure, and *Fortune* industry indicator variables for large U.S. firms that have foreign exchange rate exposure as of fiscal year-end 1990. Companies are among the 500 largest firms (by sales) in the *Fortune 500*. A firm has foreign exchange rate exposure if it has non-zero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firms on the basis of imports as a percentage of total industry sales. The second through fourth columns present the signs of the coefficient estimates based on the testable implications of Smith and Stulz (1985, S&S), Froot, Scharfstein, and Stein (1993, FSS), and DeMarzo and Duffie (1991, D&D). The full sample of 282 firms does not use the ratio of research and development expenses to sales (R&D) and contains firms for which all data are available. The partial sample of 220 firms contains firms with R&D data. $\Delta Prob.$ measures the marginal change in the probability of using derivatives resulting from a change in the independent variable. The marginal effects of the regressors on the probabilities are calculated as: $\partial Y/\partial x_i = \lambda(x'x)(1 - \lambda(x'x))x_i$, where Y = dichotomous dependent variable; x_i = i th independent variable; x = vector of independent variables; λ = the logistic cumulative distribution function; and z = vector of coefficient estimates. $\partial Y/\partial x_i$ is calculated at the means of the regressors. The t -statistics are for the logit coefficients. Summary statistics for the logit regression are presented in Panel B.

Panel A: Logit Regression Estimates									
Variable	Predicted Sign			Full Sample ($N = 282$)			Partial Sample: RD Available ($N = 220$)		
	S&S	FSS	D&D	Coeff.	$\Delta Prob.$	t -stat.	Coeff.	$\Delta Prob.$	t -stat.
Constant	NA	NA	NA	-4.634 ¹	-1.154	-2.709	-6.359 ¹	-1.550	-3.094
Managerial wealth	+	NA	NA	0.018	0.004	0.156	0.045	0.011	0.333
Managerial option ownership	-	NA	NA	0.062	0.016	0.916	0.107	0.026	1.216
Tax-loss carryforwards/total assets	+	-	NA	0.401	0.100	0.174	2.992	0.729	1.307
Long-term debt ratio (DE_AV)	+	+	NA	-1.416	-0.353	-1.172	0.132	0.032	0.086
Book-to-market ratio (BM)	-	-	-	-0.052	-0.013	-0.109	-	-	-
DE_AV*(1/BM)	+	+	+	0.561 ¹⁰	0.140	1.854	-	-	-
R&D expenses/sales (RD)	+	+	+	-	-	-	28.610 ¹	6.976	2.466
DE_AV*(RD)	+	+	+	-	-	-	-43.603	-10.632	-1.214
Quick ratio	-	-	NA	-0.919	-0.229	-1.541	-1.921 ¹	-0.468	-2.497
Number of analyst firms	NA	NA	-	0.066 ¹	0.016	2.606	0.047	0.011	1.432
Firm size	?	?	-	0.726 ¹	0.181	4.380	0.600 ¹	0.146	3.087
Pretax foreign income/total sales	+	+	+	8.671 ¹⁰	2.159	1.745	15.527 ⁵	3.786	2.028
Firm has foreign denominated debt	?	?	?	0.776 ⁵	0.193	2.434	1.239 ¹	0.302	3.086
Percentage of imports in 4-digit SIC industry	+	+	+	0.053 ¹	0.013	3.591	0.060 ¹	0.015	3.356
Firm uses other derivatives	?	?	?	0.736 ⁵	0.183	2.068	1.307 ¹	0.319	2.898
<i>Industry Indicator Variables:</i>									
Consumer goods	?	?	?	0.572	0.142	0.857	1.230	0.300	1.421
Electronics	?	?	?	-	-	-	-	-	-
Pharmaceuticals	?	?	?	-1.496 ⁵	-0.379	-0.114	-0.095	-0.406	-0.029

Table IV, Panel A: logit estimates

Table IV—Continued

Panel B: Summary Statistics for Logit Regressions						
Number of Observations	Full Sample			Partial Sample		
	Predicted Dependent Variable			Predicted Dependent Variable		
Actual dependent variable	0	1	Total	0	1	Total
0 (Discloses no use of currency derivatives)	117	31	148	77	24	101
1 (Discloses use of currency derivatives)	39	95	134	25	94	119
Total	156	126	282	102	118	220
Log likelihood at convergence	-132.41			-95.72		
Restricted log likelihood: Slopes = 0	-196.12			-151.76		

Table IV, Panel B: classification and fit

Table V
Pearson Correlation Coefficients

Pearson correlation coefficients for explanatory variables used in the logit and multinomial logit regressions (expressed in percentages). Variables are as follows: NOL is tax-loss carryforwards scaled by total assets. COV_AV is the interest coverage ratio. DE_AV is the long-term debt ratio. SIZE is the log of firm size. MGRWLTH is the log of the market value of common shares beneficially owned (excluding options) by officers and directors as a group. OPTS is the log of the market value of managerial option ownership. RD is the ratio of research and development expenses to total sales. PPE is the ratio of plant, property, and equipment investment expenditures to SIZE. BM is the book to market ratio. CONV is the ratio of convertible debt to SIZE. PS is the ratio of preferred stock to SIZE. QUICK_AV is the quick ratio. DIV_AV is the dividend yield. IO is institutional ownership. ANLST is the number of analyst firms.

	NOL	COV_AV	DE_AV	SIZE	MGRWLTH	OPTS	RD	PPE	BM	CONV	PS	QUICK_AV	DIV_AV	IO
COV_AV	-6.2%													
DE_AV	14.6 ¹	-24.7 ¹												
SIZE	-11.7 ⁵	0.7	-7.3											
MGRWLTH	-14.5 ¹	21.3 ¹	-33.9 ¹	31.6 ¹										
OPTS	-1.8	2.4	-13.1 ⁵	36.9 ¹	3.1									
RD	11.4 ¹⁰	2.2	-32.4 ¹	11.5 ⁵	11.3	15.3 ⁵								
PPE	8.7 ¹⁰	-7.5	32.9 ¹	-5.1	-30.3 ¹	-9.3 ¹⁰	-23.5 ¹							
BM	0.0	-9.1 ¹⁰	36.5 ¹	-19.2 ¹	-23.3 ¹	-13.2 ⁵	-6.5	40.0 ¹						
CONV	27.1 ¹	-6.8	17.9 ¹	-17.0 ¹	-8.5	-4.0	11.4 ¹⁰	5.4	15.2 ¹					
PS	16.8 ¹	-5.2	14.9 ¹	-3.5	-22.9 ¹	-9.6 ¹⁰	-1.4	20.3 ¹	7.1	-4.3				
QUICK_AV	3.7	11.3 ⁵	-20.9 ¹	3.4	21.7 ¹	-12.5 ⁵	30.3 ¹	-15.6 ¹	-1.4	5.2	-2.4			
DIV_AV	-6.7	6.5	21.4 ¹	-3.3	-10.7 ⁵	-7.6	-14.0 ⁵	4.0	-34.4 ¹	-6.7	-1.9	-7.0		
IO	-11.1 ⁵	-9.0 ¹⁰	-24.6 ¹	15.7 ¹	-11.4 ⁵	25.7 ¹	27.5 ¹	-5.6	-4.6	-15.7 ¹	-7.5	-11.9 ⁵	-12.3 ⁵	
ANLST	-5.3	0.2	-19.8 ¹	81.4 ¹	22.0 ¹	41.9 ¹	32.6 ¹	4.3	-15.6 ¹	-11.8 ⁵	-3.3	-6.2	-4.3	30.9 ¹

¹ (5)(10)¹ Statistically different from zero at the 1% (5%) (10%) significance level.

C. Robustness

6.4 Table VI and Table VII: instrument choice

Read:

- Table VI shows that 58.6% of exposed firms use no currency instrument, 12.1% use swaps or swap combinations, and 29.3% use forwards or forward combinations.
- Firms using forwards/forward combinations have especially high operating exposure and import competition exposure.

- Table VII estimates multinomial logits for instrument choice.
- R&D/sales and exposure variables help predict whether a firm uses swaps or forwards rather than no currency instrument.
- Firm size and use of other derivatives also predict instrument choice, consistent with economies of scale.

Economic message: Currency hedging is not one-dimensional. Firms choose instruments based partly on the source of exposure and on fixed-cost economies in risk management.

Table VI
Frequency of Use of Foreign Currency Derivative Instruments by Instrument Type

Frequency of use of selected foreign currency derivative instruments by 372 large U.S. industrial firms that have foreign exchange rate exposure as of fiscal year-end 1990. Companies are among the 500 largest firms (by sales) in the *Fortune 500*. A firm has foreign exchange rate exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firms on the basis of imports as a percentage of total industry sales. Panel A reports the frequency of use of different types of foreign currency derivatives for all firms. Panel B reports the frequency of derivatives use by type of foreign exchange rate exposure. None of the means of the proxy variables are significantly different across the choice between "Currency swaps and swap combinations" and "Currency forwards and forward combinations." The *t*-tests assume equal variances unless the null hypothesis of equal variances unless the null hypothesis of equal variances is rejected at a 10% significance level.

	No Currency Instrument		Currency Swaps and Swap Combinations		Currency Forwards and Forward Combinations		Total
Panel A: Derivative Instrument Choice							
<i>N</i>	218		45		109		372
%	58.6		12.1		29.3		100.0
Panel B: Exposure to Operating, Financing, and Competitive Risk by Derivative Instrument Choice							
	<i>N</i>	Mean	<i>N</i>	Mean	<i>N</i>	Mean	<i>N</i>
Operations							
Foreign pretax income/total sales	165	0.0171	44	0.0290 ¹⁰	105	0.0346 ¹	314
Foreign & export sales/total sales	141	0.2824	42	0.3879 ⁵	99	0.3902 ¹	282
Identifiable foreign assets/total assets	116	0.3505	39	0.3548	99	0.3432	254
Foreign debt							
Foreign long-term debt/total assets	41	0.0258	36	0.0543 ⁵	43	0.0361	120
Foreign short-term debt/total assets	25	0.0281	10	0.0255	22	0.0328	57
Competitive							
Percentage of imports in 4-digit SIC industry	218	12.03%	45	14.24%	109	17.22% ¹	372

¹ (5)(10) Significantly different from the choice of no currency instrument at the 1 (5) (10)% level, respectively.

Table VI: instrument choice and exposure

Table VII

Multinomial Logit Estimates of the Likelihood of Using Different Classes of Currency Derivatives

Multinomial logit estimates of the relation between the likelihood that a firm chooses one of two categories of derivative strategies and variables measuring hedging incentives in fiscal year 1991 for 220 large U.S. industrial firms with foreign exchange rate exposure. This sample represents all firms that have research and development (R&D) expense data. Companies are among the 500 largest firms (by sales) in the *Fortune 500*. A firm has foreign exchange rate exposure if it has nonzero foreign pretax income, positive foreign sales or debt, or is in the upper quartile of the sample firms on the basis of imports as a percentage of total industry sales (i.e., high import competition). Classes of currency derivatives choices presented in the table are expressed relative to the choice of not using currency derivatives. Choice 1 refers to using currency swaps only or currency swaps in combination with forwards, futures, and options (Swaps Only and Swap Combinations). Choice 2 refers to using foreign exchange forwards only or using foreign exchange forwards in combination with foreign exchange futures or options (Forwards Only and Forward Combinations). $\Delta Prob.$ measures the marginal change in the probability of making Choice 1 or Choice 2 resulting from a change in the independent variable, calculated at the means of the regressors. The *t*-statistics are for the coefficient estimates. Summary statistics for the multinomial logit regressions are presented in Panel B.

Panel A: Multinomial Logit Estimates						
	Derivative Instrument Choice					
	Choice 1 Swaps Only and Swap Combinations (<i>N</i> = 35)			Choice 2 Forwards Only and Forwards Combinations (<i>N</i> = 84)		
	Coefficient	$\Delta Prob.$	<i>t</i> -statistic	Coefficient	$\Delta Prob.$	<i>t</i> -statistic
Constant	-10.001 ¹	-0.623	-3.296	-3.686 ¹⁰	-0.534	-1.727
Managerial wealth	-0.337	-0.029	-1.573	0.098	0.037	0.736
Managerial option ownership	-0.098	-0.016	-0.996	0.228 ⁵	0.060	2.214
Tax loss carryforwards/total assets	0.538	-0.045	0.099	2.245	0.534	0.989
Long-term debt ratio	-3.963 ¹⁰	-0.306	-1.672	0.088	0.173	0.058
R&D expenses/total sales	12.501	-0.036	0.816	25.961 ⁵	5.968	2.377
Long-term debt ratio * R&D expenses	7.157	1.778	0.128	-32.314	-8.295	-0.929
Quick ratio	-2.886 ⁵	-0.148	-2.051	-1.905 ¹⁰	-0.363	-2.591
Analyst firms	-0.020	-0.004	-0.479	0.055 ¹⁰	0.014	1.708
Firm size	1.184 ¹	0.075	3.933	0.392 ¹⁰	0.052	1.872
Foreign pretax income/total sales	21.338 ⁵	0.955	1.926	17.658 ⁵	3.567	2.423
Long-term foreign debt/total assets	0.000	-0.000	0.408	0.001	0.000	1.106
Short-term foreign debt/total assets	-0.002	-0.000	-0.617	-0.002	-0.000	-0.363
Firm has high import competition	0.056 ⁵	0.003	2.318	0.040 ¹	0.008	2.666
Firm uses other derivatives	2.215 ¹	0.139	3.403	0.787 ¹⁰	0.111	1.789

Panel B: Summary Statistics for Multinomial Logit Regressions					
Number of Observations		Predicted Dependent Variable			
Actual dependent variable		0	1	2	Total
0 (Does not disclose the use of currency derivatives)	83	4	4	14	101
1 (Discloses using currency swaps only or currency swap combinations)	7	18	10		35
2 (Discloses using forwards only or forwards combinations)	25	8	51		84
Total	115	30	75		220
Log likelihood at convergence		-151.57			

Table VII: multinomial logit estimates

6.5 Appendix: variable definitions

Read:

- The appendix defines all variables used in the analysis.
- It clarifies timing: most flow variables are measured in the year of derivative use, while stock variables are measured at the beginning of the fiscal year.
- It explains the construction of analyst following, book-to-market, convertibles, coverage ratio, debt ratios, dividends, foreign assets, foreign debt, and foreign income variables.

Economic message: The appendix is important because the empirical tests depend heavily on mapping theoretical motives to carefully constructed proxies.

Appendix

This appendix provides a summary of all explanatory variables used in our analysis and a detailed description of the method of calculation. The explanatory variables are generally constructed as follows: Flow variables are measured during the year of derivatives use (1991), and stock variables are measured at the beginning of the year (fiscal year end 1990). Implicit in this construction are two assumptions: First, decisions to use derivatives (measured at fiscal year-end 1991) are based on information available to the firm during 1991; second, actual flows in 1991 are the best proxy for management's expectations of 1991. Flow variables for 1990, however, may be a better proxy for management's expectations depending on the timing of the flow, the timing of the decision to use derivatives, and the degree to which management uses past performance to estimate future flows.

Variable Name	Variable Description
ANLST	The number of analyst firms covering a company as of September/October 1990, according to <i>Nelson's Directory of Investment Research</i> . Alternatively, ANLST is measured as the number of analysts, rather than firms, following each sample firm. The results are qualitatively similar using both specifications of ANLST, so we present only those based on the number of firms.
BM	Ratio of book-to-market value of the firm. Book value of common shareholders' equity as of the end of fiscal year 1990 is total assets less total liabilities less outstanding preferred stock (COMPUSTAT data items 6, 181, and 130, respectively). Market value is closing share price times common shares outstanding at year-end 1990 (COMPUSTAT data items 199 and 25, respectively).
CONV	Ratio of book value of total convertible debt as of fiscal year-end 1990 (COMPUSTAT data item 79) to SIZE.
COV ^a	Interest coverage ratio. Ratio of pretax income for 1991 (COMPUSTAT data item 170) plus interest expense for 1991 (COMPUSTAT data item 15) to interest expense plus capitalized interest (COMPUSTAT data item 239) for 1991.
DE ^a	Long-term debt ratio. Ratio of book value of long-term debt as of the end of fiscal year 1990 (COMPUSTAT data items 34 plus 9) to SIZE.
DIV ^a	Dividend yield. Ratio of cash dividend per share in 1991 (COMPUSTAT data item 26) to closing price per share as of fiscal year-end 1990 (COMPUSTAT data item 199).
FORASSETS	Ratio of identifiable foreign assets at year-end 1990 (annual report) to total assets.
FORDEBT	An indicator variable equal to one if the firm has either nonzero LTFRDT or STFRDT or qualitatively discusses the existence of foreign-denominated debt in its annual report.

Appendix variable definitions, part 1

Variable Name	Variable Description
FORSALES	Ratio of foreign sales plus export sales for the year ended 1991 (annual report) to sales (COMPUSTAT data item 12).
IMPORTS	The percentage of imports to total production (domestic production plus imports) as of December 31, 1991. These data are reported by the Department of Commerce for 4-digit SIC codes and matched to the sample based on COMPUSTAT dnum. Where 4-digit dnums are not available, we obtain 4-digit industry classifications from the 1991 annual report.
IO	The percentage of the firm's common shares outstanding held by institutional owners at fiscal year-end 1990 as reported by <i>Nelson's Directory of Investment Research</i> . Institutional investors include banks, brokerage firms, investment managers, insurance companies, mutual funds, and pension funds. Similar results related to institutional ownership are obtained using figures reported by <i>Business Week</i> and <i>O'Neil Database</i> .
LTFRDT	The long-term and current portion of debt issued with an original maturity greater than one year, denominated in a currency other than US dollars (annual report) at year-end 1990.
MGRWLTH	Market value of common shares beneficially owned (excluding shares that can be obtained within 60 days through the exercise of options) by officers and directors as a group as of fiscal year-end 1990 reporting date (Compact Disclosure, proxy statements, and 10-K filings).
NOL	The portion of prior and current year net operating losses applied as a reduction of taxable income in 1990 (COMPUSTAT data item 52) scaled by total assets (COMPUSTAT data item 6). The book value of NOL is only reported on COMPUSTAT if it appears in the firm's tax footnote. When net operating loss carryforwards are missing, we examine the firm's annual reports and 10-Ks to determine the book value of NOL if available, or the tax value of net operating loss carryforwards if the book value is unavailable.
OPTS	The log of the market value of shares that can be obtained by exercise of options. The options are exercisable within 60 days of the reporting date for common shares beneficially owned by officers and directors as a group (Compact Disclosure, proxy statements, and 10-K filings).
OTHDERIV	An indicator variable equal to one if the firm discloses using either interest-rate-based or commodity-based derivative instruments; equal to zero otherwise.
PPE	Intensity of capital investment. Ratio of property, plant, and equipment at year-end 1990 (COMPUSTAT data item 187) to SIZE.
PS	Ratio of book value of total preferred stock as of the end of fiscal year 1990 (COMPUSTAT data item 130) to SIZE.
QUICK ^a	Quick ratio. Ratio of cash and short-term investments as of fiscal year-end 1990 (COMPUSTAT data item 1) to current liabilities as of fiscal year-end 1990 (COMPUSTAT data items 34, 70, 71, and 72).
RD	Ratio of R&D expenses during 1991 (COMPUSTAT data item 46) to sales (COMPUSTAT data item 12).
SIZE	Market value of the firm at fiscal year-end 1990. Natural logarithm of the sum of the market value of equity (COMPUSTAT data item 199 times COMPUSTAT data item 25), book value of long-term debt (COMPUSTAT data items 9 and 34), and book value of preferred stock (COMPUSTAT data item 130).
STFRDT	Debt issued with an original maturity less than one year denominated in a currency other than U.S. dollars (annual report) at year-end 1990.

Appendix variable definitions, part 2

7 Result synthesis

7.1 Support for underinvestment-based hedging

The strongest evidence supports the Froot-Scharfstein-Stein style view: hedging is valuable when external finance is costly and internal cash flow volatility can force the firm to cut valuable investment. R&D-intensive firms and firms with tighter liquidity are more likely to use currency derivatives.

Interpretation: This result is important because it treats derivatives as an investment-support tool rather than a pure risk-minimization device.

7.2 Support for exposure-based motives

Firms with foreign income, foreign sales, foreign debt, and high import competition are more likely to use currency derivatives. This supports the basic idea that hedging demand rises with the magnitude of exposure.

Interpretation: Exposure alone is not enough to explain hedging, but exposure is necessary: the benefits of hedging are larger when cash flows or balance sheets are more sensitive to exchange rates.

7.3 Evidence on hedging costs and scale economies

Larger firms and firms already using other derivatives are more likely to use currency derivatives. This is consistent with fixed costs of setting up derivative programs, such as treasury expertise, legal/accounting systems, and dealer relationships.

Interpretation: Hedging is not free. The decision depends on both benefits and the fixed and variable costs of implementing risk management.

7.4 Substitution between debt and derivatives

For firms with foreign operations and foreign-denominated debt, R&D and liquidity variables are less important predictors of currency derivative use. This suggests that foreign-denominated debt can naturally hedge foreign operations.

Interpretation: The paper recognizes that firms can hedge with instruments other than derivatives. This is important for interpreting nonuse of currency derivatives.

8 Short summary

- The paper studies 372 Fortune 500 nonfinancial firms with foreign exchange exposure.
- About 41% of these firms use currency derivatives.
- Firms with more growth opportunities and tighter financial constraints are more likely to use currency derivatives.
- This supports theories in which hedging protects investment by stabilizing internal funds.
- Firms with greater foreign exchange exposure are also more likely to hedge.
- Larger firms and firms already using other derivatives are more likely to use currency derivatives, supporting the importance of scale economies.
- Foreign-denominated debt can substitute for currency derivatives as a natural hedge.
- Instrument choice differs by exposure source and firm characteristics.
- The evidence is more consistent with hedging than with speculation.

9 Conclusion

9.1 Core conclusion

Geczy, Minton, and Schrand show that large U.S. firms use currency derivatives for systematic economic reasons. Derivative use is strongest among firms with both higher potential benefits from hedging and lower costs of maintaining a hedging program.

9.2 Economic intuition

Currency risk matters most when it can disrupt investment. Firms with high R&D and other growth opportunities are especially vulnerable to cash-flow shocks if external finance is costly. Hedging currency risk preserves internal funds and helps the firm continue investing. Firms also hedge more when they have greater foreign exchange exposure and when scale economies make hedging cheaper.

9.3 Incremental takeaway

The paper's enduring contribution is that it links observed derivative disclosures to a theory-based set of corporate hedging motives. It shows that derivatives are not merely speculative tools or treasury side products. For many firms, they appear to be part of a broader policy to manage financing frictions and protect investment opportunities.